TARS4 — wiring guide

The 110 TARS wires (33 in 26 AWG, the rest in 30 AWG), in build order. Each step is tested on the bench before the next; everything is done outside TARS, fitting comes at the end. The board plans (renders/ tars4_board_*.svg) give hole by hole, this guide gives both ends of every wire.

WIRE END CODE (THE SAME ON THE BOARD PLANS)

● **Soldered**: on a board hole (plan shown) or on a module pad / lead. Tin the wire, solder, heat-shrink over the joint if it is exposed.

DF **Dupont, female contact** crimped (SN-28B, 30 AWG folded in two; two wires in one contact: together, without folding), slid into housing Xn at the cavity shown (cavity 1 = pin 1, paint dot).

DM **Dupont, male contact**: the module side of an inline pair (pigtails of X3, X7, X15).

XHF **JST-XH female** crimped, on the power block side (X10, X13, X14) or the cell side (X12).

XHM **JST-XH male** crimped, the other side of the pair.



Wire colour, thick = 26 AWG, strokes = black heat-shrink rings at both ends (1 = I2S / PIR, 2 = I2C, 3 = microSD).



Power pairs: red / black = battery and 5 V, white / blue = regulated 3V3. Rails sharing red / black carry a **coloured band** (10 mm of coloured heat-shrink, shrunk 2–3 cm from each end, the same on the + and on its GND): none = cell B+ / B−, **white** = TP4056 OUT+ / OUT−, **yellow** = VPATH, **orange** = VSYS, **purple** = VUSB. Slip the band on before soldering or crimping.

DIAGRAM BLOCKS (FRAME COLOUR)

P **Hand-wired board**: P1, P2 (stripboards), P3 S3 carrier, P4 power path (pad perfboards); plans renders/ tars4_board_*.svg.

M **Module or part** under its name: TP4056, TPS63020, INMP441, AM312, microSD reader, switch, cells, speaker...

X **Dupont connector**, one side (female or male); cavities are numbered from pin 1.

X **JST-XH connector**, one side (female or male).

In the diagrams, TARS is seen from the front (left leg on the left), one column per TARS column, the tubes between them. A block = a board, a module or one side of a connector; each line = one wire end. A wire started at one step and finished later leaves the block with an arrow "→ st. n".

1 Power block, on the bench

left leg: TP4056, power-path board, TPS63020 soldered together · 22 wires finished · housings to fill: X10 F, X12 M, X13 F, X14 F, X16 M

Before

1. Power-path board: parts, bridges and GND bus per renders/tars4_board_power.svg (Q1 / Q5 with wire-wrap wire).
2. TP4056: R_PROG replaced by 2.4 k, then tested alone: 3.7 V bench supply on B+ / B−, current < 10 µA. Polarities: C9 + on IN+, D3 band (cathode) on IN+ (reversed, it shorts the 5 V). TPS63020: EN tied to VIN, PS/SYNC to GND.
3. Crowded pads (joints ≥ 0.65 mm from the board end): splice 1 cm before the pad the two VUSB wires of IN+ (to X10 and to the power-path board), the two B+ (to X12 and X16, with the B+ sense wire to the power-path board) and the two B−: a single conductor per pad, next to C9 / D3.
4. Solder the wires below (4–5 cm loops between the three boards), then crimp the XH contacts and slide them into X10, X12, X13 and X14.
5. X12 and X16 on the block side (M): while unplugged, capped with an empty female XH housing. Their pins carry B+ / B− as soon as a cell is plugged in, until step 7.

Bench test

- ☐ 3.7 V bench supply on X12 (block side, M): OUT+ ≈ B+; current without USB noted (D1 leakage adds to the TP4056 one: a few tens of µA possible; power_budget.md).
- ☐ X10 cavities 3–4 bridged (switch simulated): VSYS ≈ B+; 3.3 V on X13 cavity 3; VBAT/2 ≈ B+ / 2 on X14 cavity 1.
- ☐ 5 V on X10 cavities 1–2: charge LED, USB_DET ≈ 3 V on X14 cavity 2, Q1 off.

Step 1 · Power block, on the bench: diagram

Wires finished at this step (22)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W1	red 26 AWG, purple band	VUSB	**P4** power path · hole F8 (wire 1 of the plan)	TP4056 · IN+ (VUSB)	—	**10** → 13
W2	red 26 AWG, white band	OUT+	**P4** power path · hole B4 (wire 2 of the plan)	TP4056 · OUT+	—	**11** → 14
W3	black 26 AWG, white band	GND	**P4** power path · hole C1 (wire 7 of the plan)	TP4056 · OUT−	—	**11** → 14
W4	red 30 AWG	B+	**P4** power path · hole J8 (wire 10 of the plan)	TP4056 · B+ (sense)	—	**11** → 14
W5	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 5 of the plan)	TPS63020 · VIN	—	**12** → 15
W6	black 26 AWG, orange band	GND	**P4** power path · hole D1 (wire 8 of the plan)	TPS63020 · GND	—	**12** → 15
W7	white 30 AWG	3V3	**P4** power path · hole H8 (wire 13 of the plan)	TPS63020 · VOUT	—	**12** → 15
W8	red 26 AWG, yellow band	VPATH	**P4** power path · hole A8 (wire 3 of the plan)	XHF **X10** J1 + SW1 · case 3 VPATH	—	**11** → 14
W9	red 26 AWG, orange band	VSYS	**P4** power path · hole D3 (wire 4 of the plan)	XHF **X10** J1 + SW1 · case 4 VSYS	—	**10** → 13
W10	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 6 of the plan)	XHF **X13** power block outputs · case 1 VSYS	—	**11** → 14
W11	black 26 AWG, orange band	GND	**P4** power path · hole E1 (wire 9 of the plan)	XHF **X13** power block outputs · case 2 GND amp	—	**11** → 14
W12	brown 30 AWG	VBAT/2	**P4** power path · hole J3 (wire 11 of the plan)	XHF **X14** measurements · case 1 VBAT/2	—	**10** → 13
W13	orange 30 AWG	USB_DET	**P4** power path · hole E2 (wire 12 of the plan)	XHF **X14** measurements · case 2 USB_DET	—	**11** → 14
W14	brown 30 AWG, 3 rings	SD_EN	**P4** power path · hole H5 (wire 15 of the plan)	XHF **X14** measurements · case 3 SD_EN	—	**9** → 12
W15	red 26 AWG, purple band	VUSB	TP4056 · IN+	XHF **X10** J1 + SW1 · case 1 VUSB	—	**9** → 12
W16	black 26 AWG, purple band	GND	TP4056 · IN−	XHF **X10** J1 + SW1 · case 2 GND	—	**9** → 12
W17	red 26 AWG	B+	TP4056 · B+	XHM **X12** left cell · case 1 B+ cell junction: two wires soldered on the B+ pad	—	**10** → 13
W18	black 26 AWG	B−	TP4056 · B−	XHM **X12** left cell · case 2 B− cell junction: two wires soldered on the B− pad	—	**10** → 13
W19	red 26 AWG	B+	TP4056 · B+	XHM **X16** right cell · case 1 B+ to X16, parked above the power-path board	—	**13** → 16
W20	black 26 AWG	B−	TP4056 · B−	XHM **X16** right cell · case 2 B−	—	**13** → 16
W21	white 26 AWG	3V3	TPS63020 · VOUT	XHF **X13** power block outputs · case 3 3V3 along the logic GND wire	—	**11** → 14
W22	blue 26 AWG	GND	TPS63020 · GND	XHF **X13** power block outputs · case 4 GND log logic GND, along the 3V3	—	**11** → 14

Wires started here, finished later (1)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut

W58	**1→5**	white 30 AWG, 3 rings	SD_3V3	**P4** power path · hole I6 (wire 14 of the plan)	DF	**X15** microSD · case 4 3V3	—	**16** → **19**

2 Charge port, switch, cells

the foot and cell cables up to X10, X12 and X16 · 6 wires finished · housings to fill: X10 M, X12 F

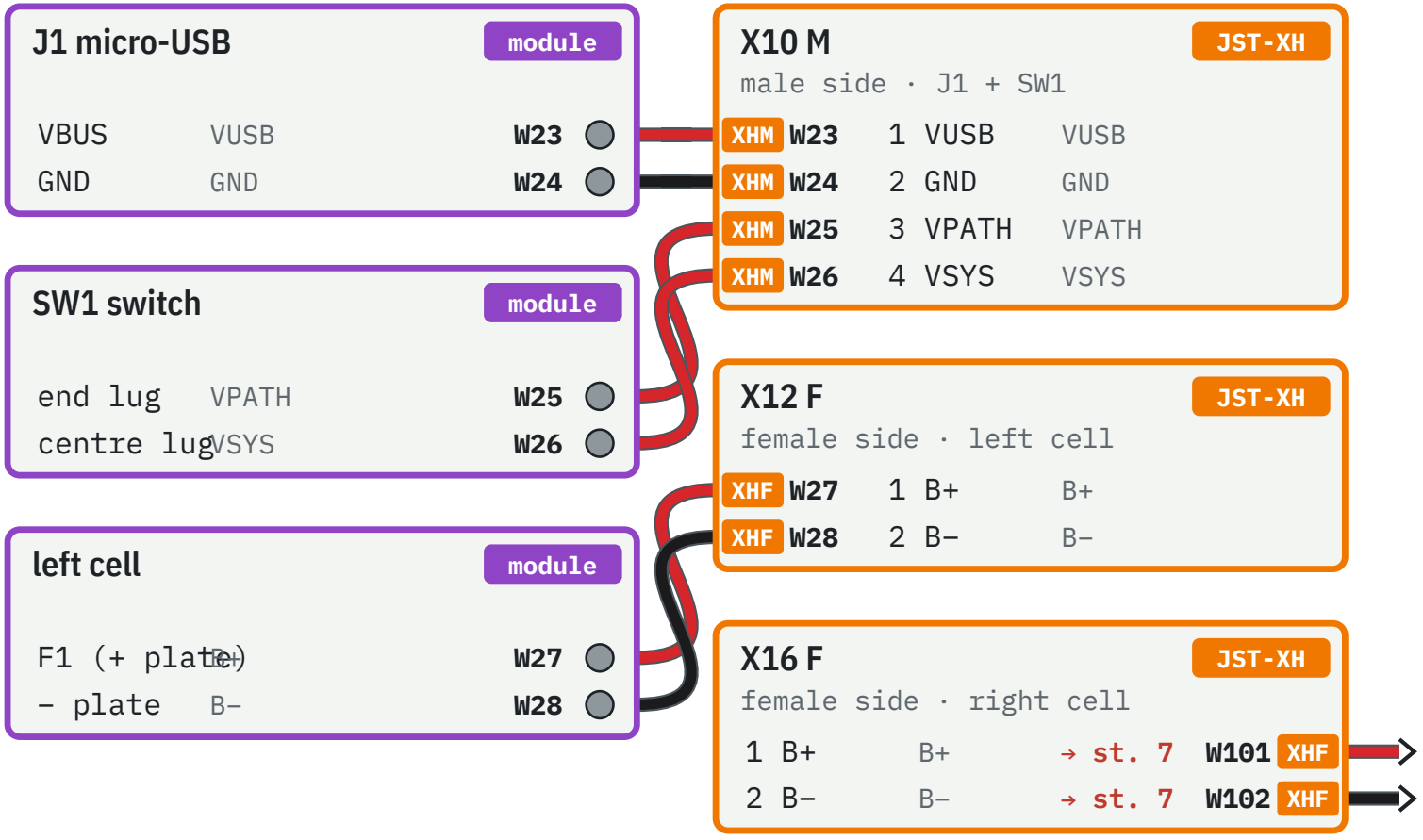
Before

1. Solder F1 on the left + plate (F2 on the right one at step 7). Never solder on a cell.
2. Right cell wires: long enough to cross the three tubes (length from the table, already + 3 cm, offered up on TARS before cutting). Crimp X16 (female) now; the right cell end stays bare.

Bench test

- ☐ Left cell alone: X12 plugged in, switch on, VSYS ≈ B+.
- ☐ Charger on J1: charge LED; unplugged: VUSB < 0.5 V within ~1 s.

LEFT LEG


J1 micro-USB
module
VBUS VUSB W23
GND GND W24
SW1 switch
module
end lug VPATH W25
centre lug VSYS W26
left cell
module
F1 (+ plate) B+ W27
− plate B− W28
X10 M
JST-XH
male side · J1 + SW1
XHM W23 1 VUSB VUSB
XHM W24 2 GND GND
XHM W25 3 VPATH VPATH
XHM W26 4 VSYS VSYS
X12 F
JST-XH
female side · left cell
XHF W27 1 B+ B+
XHF W28 2 B− B−
X16 F
JST-XH
female side · right cell
1 B+ B+ → st. 7 W101 XHF
2 B− B− → st. 7 W102 XHF

Wires finished at this step (6)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W23	 red 26 AWG, purple band	VUSB	● J1 micro-USB · VBUS	XHM **X10** J1 + SW1 · case 1 VUSB twisted with the GND	—	**24** → 27
W24	 black 26 AWG, purple band	GND	● J1 micro-USB · GND	XHM **X10** J1 + SW1 · case 2 GND	—	**24** → 27
W25	 red 26 AWG, yellow band	VPATH	● SW1 switch · end lug	XHM **X10** J1 + SW1 · case 3 VPATH	—	**24** → 27
W26	 red 26 AWG, orange band	VSYS	● SW1 switch · centre lug	XHM **X10** J1 + SW1 · case 4 VSYS	—	**24** → 27
W27	 red 26 AWG	B+	● left cell · F1 (+ plate)	XHF **X12** left cell · case 1 B+ twisted with B–	—	**18** → 21
W28	 black 26 AWG	B–	● left cell · – plate	XHF **X12** left cell · case 2 B–	—	**24** → 27

Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W101	2→7	 red 26 AWG	B+	● right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B–	left tube + middle tube + right tube	**52** → 55
W102	2→7	 black 26 AWG	B–	● right cell · – plate	XHF **X16** right cell · case 2 B– as B+; twisted with B+	left tube + middle tube + right tube	**57** → 60

3 S3 carrier

the 27 controller wires, soldered on the carrier side · 0 wires finished · housings to fill: none

Before

1. Plan renders/tars4_board_s3.svg: 2 × 8 headers (V contact removed, headers glued), standing parts, bridges, GND bus, jumpers.
2. Solder the 27 wires on the carrier side only, long enough (6–8 cm service loop + route). The other end stays bare: it is finished at the step shown.

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 removed: each wire ↔ its pin with the ohmmeter, GND isolated from 3V3 and the signals.
- ☐ S3 plugged in, PC USB: test program flashed, the S3 boots.

Step 3 · S3 carrier: diagram

Wires finished at this step (0)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut

Wires started here, finished later (27)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W29	3→4	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → 31
W30	3→4	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → 33
W31	3→4	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → 29
W32	3→4	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → 28
W33	3→4	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → 30
W34	3→4	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → 29
W35	3→4	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → 39
W36	3→4	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → 38
W37	3→4	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → 25
W38	3→4	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → 25
W39	3→4	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → 25
W40	3→4	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → 27
W41	3→4	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	—	**22** → 25
W42	3→4	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	—	**23** → 26
W59	3→5	brown 30 AWG	VBAT/2	**P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**49** → 52
W60	3→5	orange 30 AWG	USB_DET	**P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**52** → 55
W61	3→5	brown 30 AWG, 3 rings	SD_EN	**P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → 55
W62	3→5	purple 30 AWG, 3 rings	SD_CMD	**P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → 60
W63	3→5	green 30 AWG, 3 rings	SD_D0	**P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → 60
W64	3→5	yellow 30 AWG, 3 rings	SD_CLK	**P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → 58
W73	3→6	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → 52
W74	3→6	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → 54
W75	3→6	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → 50
W76	3→6	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → 49
W77	3→6	green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	DF **X5** MAX98357A · case 3 DIN	middle tube	**57** → 60

W78	**3→6**	purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	**DF**	**X5** MAX98357A · case 1 LRC	middle tube	**58** → **61**
W79	**3→6**	yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	**DF**	**X5** MAX98357A · case 2 BCLK	middle tube	**59** → **62**

4 P1 and the left middle

screen 1, mic, accelerometer · 29 wires finished · housings to fill: X1 F, X2 F, X3 F, X3 M

Before

1. Plan renders/tars4_board_p1.svg. 1 × 8 right-angle header on the back of screen 1 (X1). ADXL345: straight header, INT2 removed and its cavity plugged (X2).
2. INMP441: 5-wire pigtail (male contacts, X3); L/R bridged to GND on its board.
3. X2 SDA and SCL: double contact, crimp together (without folding back) the S3 wire and the one leaving for P2 (bare end, soldered at step 6).

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 on USB, switch off: screen 1 displays, the mic gives I2S samples, the ADXL345 answers at 0x53 and INT1 toggles.

Step 4 · P1 and the left middle: diagram

Wires finished at this step (29)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W29	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → 31
W30	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → 33
W31	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → 29
W32	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → 28
W33	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → 30
W34	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → 29
W35	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → 39
W36	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → 38
W37	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → 25
W38	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → 25
W39	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → 25
W40	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → 27
W41	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	—	**22** → 25
W42	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	—	**23** → 26
W43	yellow 30 AWG, 1 ring	MIC_BCLK	INMP441 · SCK	DM **X3** INMP441 · case 1 SCK 5–6 cm pigtail	—	**6** → 7
W44	purple 30 AWG, 1 ring	MIC_WS	INMP441 · WS	DM **X3** INMP441 · case 2 WS	—	**6** → 7
W45	green 30 AWG, 1 ring	MIC_SD	INMP441 · SD	DM **X3** INMP441 · case 3 SD	—	**6** → 7
W46	white 30 AWG	3V3	INMP441 · VDD	DM **X3** INMP441 · case 4 VDD	—	**6** → 7
W47	blue 30 AWG	GND	INMP441 · GND	DM **X3** INMP441 · case 5 GND L/R bridged to GND on the board	—	**6** → 7
W48	blue 30 AWG	GND	**P1** left middle distribution · hole A10 (wire 1 of the plan)	DF **X1** screen 1 · case 1 GND	—	**15** → 18
W49	white 30 AWG	3V3	**P1** left middle distribution · hole A9 (wire 7 of the plan)	DF **X1** screen 1 · case 2 VCC	—	**15** → 18
W50	brown 30 AWG	BL_EN	**P1** left middle distribution · hole B8 (wire 13 of the plan)	DF **X1** screen 1 · case 8 BLK	—	**15** → 18
W51	red 30 AWG	DC	**P1** left middle distribution · hole B7 (wire 16 of the plan)	DF **X1** screen 1 · case 6 DC	—	**15** → 18
W52	orange 30 AWG	RST	**P1** left middle distribution · hole B6 (wire 19 of the plan)	DF **X1** screen 1 · case 5 RES	—	**15** → 18
W53	green 30 AWG	MOSI	**P1** left middle distribution · hole B5 (wire 22 of the plan)	DF **X1** screen 1 · case 4 SDA	—	**16** → 19
W54	blue 30 AWG	GND	**P1** left middle distribution · hole D11 (wire 6 of the plan)	DF **X3** INMP441 · case 5 GND	—	**16** → 19
W55	white 30 AWG	3V3	**P1** left middle distribution · hole D9 (wire 10 of the plan)	DF **X3** INMP441 · case 4 VDD	—	**15** → 18

W56	white 30 AWG	3V3	DF **X3** INMP441 · case 4 VDD	DF **X2** ADXL345 · case 2 VCC double contact in X3 (with the wire from P1)	–	**12** → 15
W57	blue 30 AWG	GND	DF **X3** INMP441 · case 5 GND	DF **X2** ADXL345 · case 1 GND double contact in X3 (with the wire from P1)	–	**12** → 15

Wires started here, finished later (11)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W65	4→5	blue 26 AWG	GND	● **P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → 36
W66	4→5	white 26 AWG	3V3	● **P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → 36
W67	4→5	blue 30 AWG, 3 rings	GND	● **P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → 38
W80	4→6	blue 30 AWG	GND	● **P1** left middle distribution · hole C11 (wire 5 of the plan)	● **P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → 32
W81	4→6	white 30 AWG	3V3	● **P1** left middle distribution · hole F9 (wire 11 of the plan)	● **P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → 32
W82	4→6	brown 30 AWG	BL_EN	● **P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → 30
W83	4→6	red 30 AWG	DC	● **P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → 30
W84	4→6	orange 30 AWG	RST	● **P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → 30
W85	4→6	green 30 AWG	MOSI	● **P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → 31
W86	4→6	green 30 AWG, 2 rings	SDA	● **P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → 39
W87	4→6	yellow 30 AWG, 2 rings	SCL	● **P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → 40

5 Left tube: power block ↔ left middle, microSD

everything crossing the left tube · 15 wires finished · housings to fill: X13 M, X14 M, X15 F, X15 M

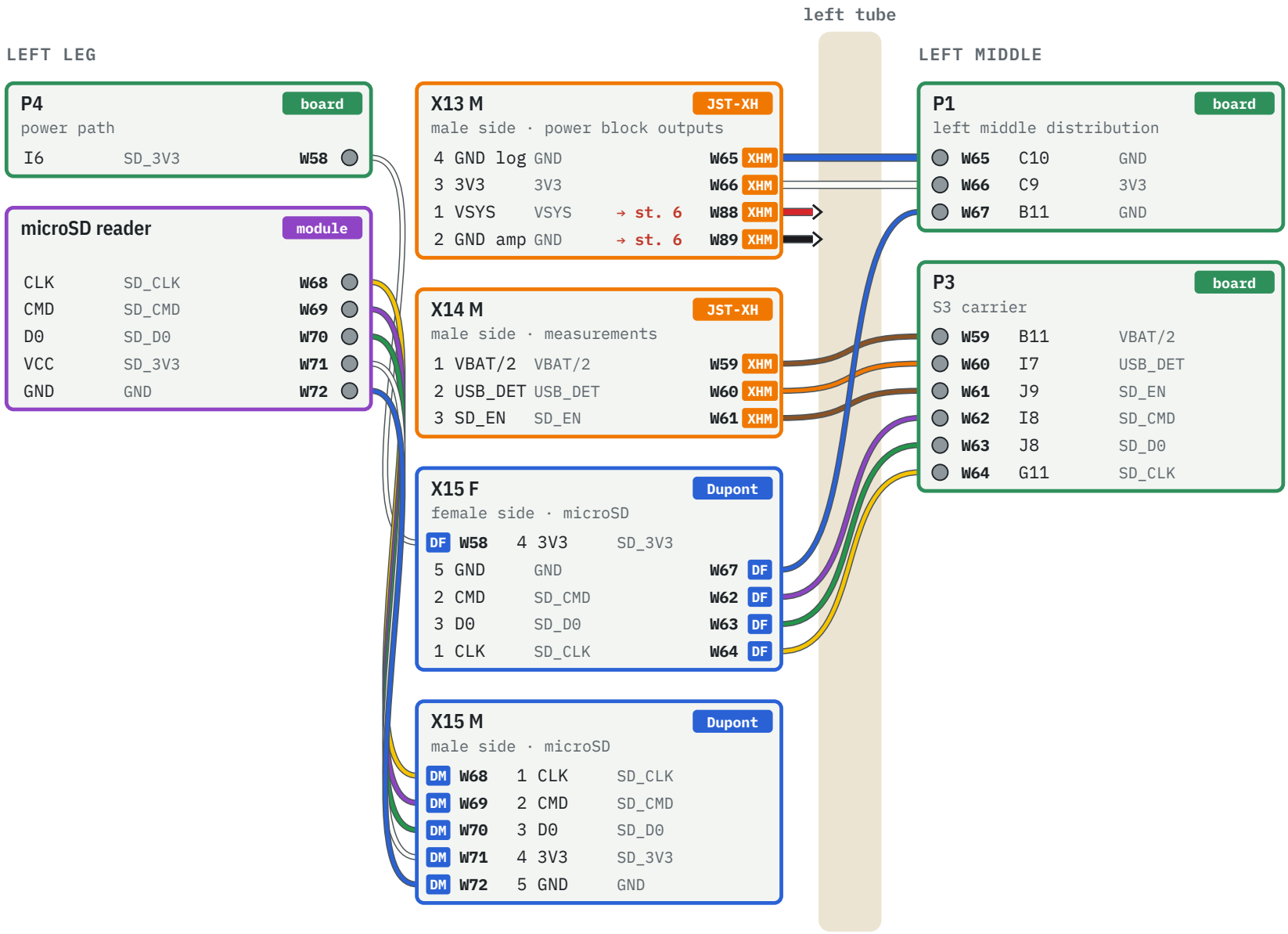
Before

1. Thread the left brass tube onto the listed wires BEFORE crimping the far end. The amplifier pair (step 6) and the right cell pair (step 2) also cross this tube: thread them at the same time.
2. microSD reader: C4 10 µF tantalum lying flat (+ on VCC; only ~6.6 mm up to the TPS63020) and C5 100 nF on its VCC / GND pins; without pull-ups on the reader, R5–R9 10 k from CMD, D0, D1, D2 and D3 to VCC (SD_3V3). Then a 5-wire pigtail (male contacts, X15).

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Without USB, switch on: the S3 boots from the cells (3.3 V on the carrier).
- ☐ VBAT/2 and USB_DET read by the S3; the microSD is read; SD_EN high cuts its 3.3 V.

Step 5 · Left tube: power block ⇔ left middle, microSD: diagram



Wires finished at this step (15)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W58	white 30 AWG, 3 rings	SD_3V3	**P4** power path · hole I6 (wire 14 of the plan)	DF **X15** microSD · case 4 3V3	—	**16** → 19
W59	brown 30 AWG	VBAT/2	**P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**49** → 52
W60	orange 30 AWG	USB_DET	**P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**52** → 55
W61	brown 30 AWG, 3 rings	SD_EN	**P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → 55
W62	purple 30 AWG, 3 rings	SD_CMD	**P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → 60
W63	green 30 AWG, 3 rings	SD_D0	**P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → 60
W64	yellow 30 AWG, 3 rings	SD_CLK	**P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → 58
W65	blue 26 AWG	GND	**P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → 36
W66	white 26 AWG	3V3	**P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → 36
W67	blue 30 AWG, 3 rings	GND	**P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → 38
W68	yellow 30 AWG, 3 rings	SD_CLK	microSD reader · CLK	DM **X15** microSD · case 1 CLK 5–6 cm pigtail	—	**6** → 7
W69	purple 30 AWG, 3 rings	SD_CMD	microSD reader · CMD	DM **X15** microSD · case 2 CMD	—	**6** → 7
W70	green 30 AWG, 3 rings	SD_D0	microSD reader · D0	DM **X15** microSD · case 3 D0	—	**6** → 7
W71	white 30 AWG, 3 rings	SD_3V3	microSD reader · VCC	DM **X15** microSD · case 4 3V3	—	**6** → 7
W72	blue 30 AWG, 3 rings	GND	microSD reader · GND	DM **X15** microSD · case 5 GND	—	**6** → 7

Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W88	5→6	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → 51
W89	5→6	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes	left tube + middle tube	**48** → 51

6 Middle tube and right middle

screen 2, amplifier, speaker, AM312, P2 · 28 wires finished · housings to fill: X2 F, X4 F, X5 F, X6 F, X7 F, X7 M, X13 M

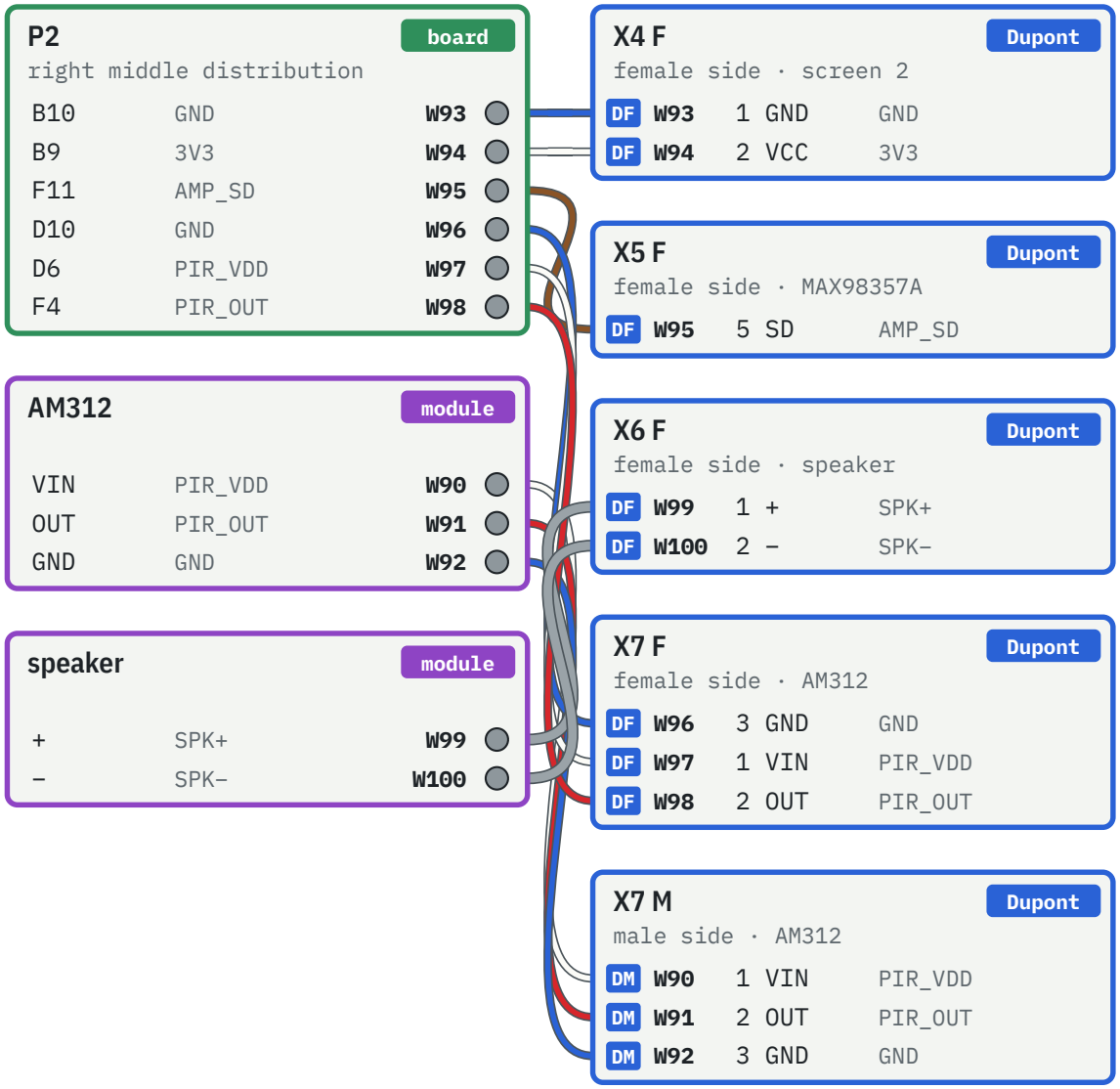
Before

1. Plan renders/tars4_board_p2.svg. Right-angle header on the back of screen 2 (X4). Amplifier: 1 × 4 (GAIN removed) + 1 × 3 headers (X5), 2-pin header instead of the screw terminal (X6), C7 + on VIN, band (–) to GND.
2. AM312: 3-wire pigtail (male contacts, X7), breakout pins trimmed flush, breakout slipped into a heat-shrink sock (it passes 5 mm from the middle tube exit). Thread the middle tube before crimping.

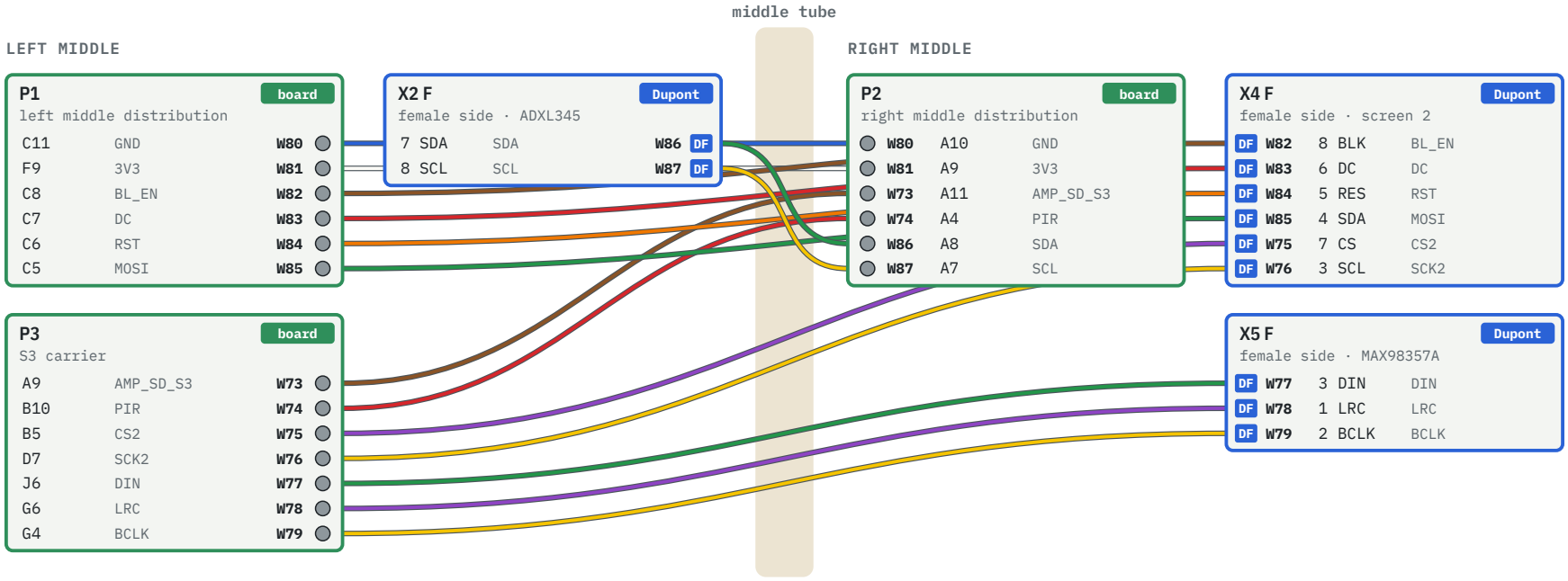
Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Screen 2 displays; sound in the speaker; AMP_SD low cuts the amplifier; PIR rises when a hand passes.

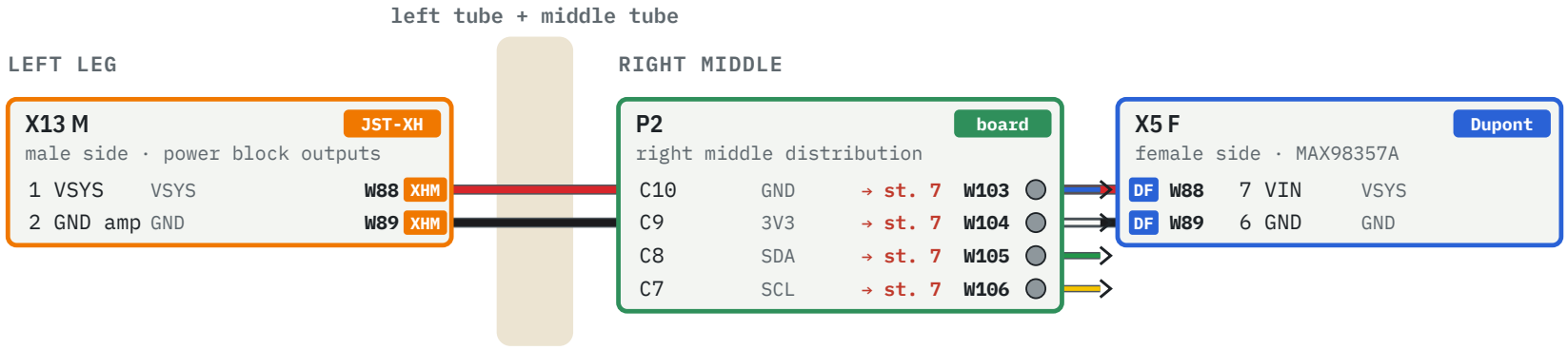
RIGHT MIDDLE



Step 6 · Middle tube and right middle: diagram 2/3: wires through middle tube



Step 6 · Middle tube and right middle: diagram 3/3: wires through right tube, through left tube + middle tube



Wires finished at this step (28)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W73	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → 52
W74	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → 54
W75	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → 50
W76	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → 49
W77	green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	DF **X5** MAX98357A · case 3 DIN	middle tube	**57** → 60
W78	purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	DF **X5** MAX98357A · case 1 LRC	middle tube	**58** → 61
W79	yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	DF **X5** MAX98357A · case 2 BCLK	middle tube	**59** → 62
W80	blue 30 AWG	GND	**P1** left middle distribution · hole C11 (wire 5 of the plan)	**P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → 32
W81	white 30 AWG	3V3	**P1** left middle distribution · hole F9 (wire 11 of the plan)	**P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → 32
W82	brown 30 AWG	BL_EN	**P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → 30
W83	red 30 AWG	DC	**P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → 30
W84	orange 30 AWG	RST	**P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → 30
W85	green 30 AWG	MOSI	**P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → 31
W86	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → 39
W87	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → 40
W88	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → 51
W89	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes	left tube + middle tube	**48** → 51
W90	white 30 AWG	PIR_VDD	AM312 · VIN	DM **X7** AM312 · case 1 VIN 5–6 cm pigtail	—	**6** → 7
W91	red 30 AWG, 1 ring	PIR_OUT	AM312 · OUT	DM **X7** AM312 · case 2 OUT	—	**6** → 7
W92	blue 30 AWG	GND	AM312 · GND	DM **X7** AM312 · case 3 GND	—	**6** → 7
W93	blue 30 AWG	GND	**P2** right middle distribution · hole B10 (wire 4 of the plan)	DF **X4** screen 2 · case 1 GND	—	**14** → 17

W94	white 30 AWG	3V3	**P2** right middle distribution · hole B9 (wire 8 of the plan)	DF **X4** screen 2 · case 2 VCC	—	**15** → **18**
W95	brown 30 AWG, 1 ring	AMP_SD	**P2** right middle distribution · hole F11 (wire 2 of the plan)	DF **X5** MAX98357A · case 5 SD	—	**16** → **19**
W96	blue 30 AWG	GND	**P2** right middle distribution · hole D10 (wire 6 of the plan)	DF **X7** AM312 · case 3 GND	—	**14** → **17**
W97	white 30 AWG	PIR_VDD	**P2** right middle distribution · hole D6 (wire 14 of the plan)	DF **X7** AM312 · case 1 VIN	—	**15** → **18**
W98	red 30 AWG, 1 ring	PIR_OUT	**P2** right middle distribution · hole F4 (wire 16 of the plan)	DF **X7** AM312 · case 2 OUT	—	**17** → **20**
W99	grey 26 AWG	SPK+	speaker · +	DF **X6** speaker · case 1 + + marked with a pen at both ends	—	**11** → **14**
W100	grey 26 AWG	SPK–	speaker · –	DF **X6** speaker · case 2 –	—	**11** → **14**

Wires started here, finished later (4)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W103	**6→7**	blue 30 AWG	GND	**P2** right middle distribution · hole C10 (wire 5 of the plan)	DF **X9** BME280 · case 2 GND	right tube	**27** → **30**
W104	**6→7**	white 30 AWG	3V3	**P2** right middle distribution · hole C9 (wire 9 of the plan)	DF **X9** BME280 · case 1 VCC	right tube	**27** → **30**
W105	**6→7**	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole C8 (wire 11 of the plan)	DF **X9** BME280 · case 4 SDA	right tube	**27** → **30**
W106	**6→7**	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole C7 (wire 13 of the plan)	DF **X9** BME280 · case 3 SCL	right tube	**28** → **31**

7 Right tube and right leg

DS3231, BME280, right cell · 10 wires finished · housings to fill: X8 F, X9 F, X16 F

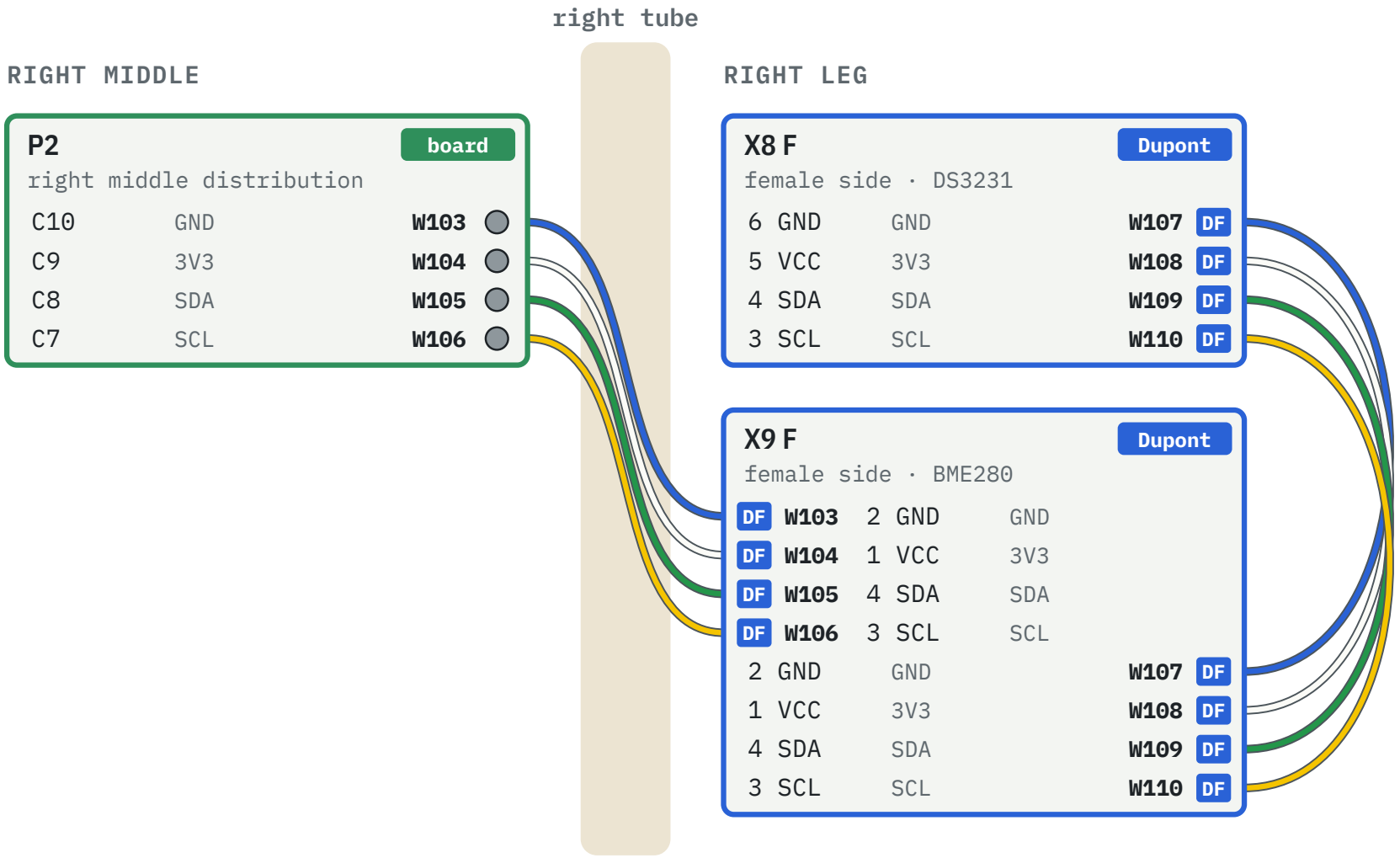
Before

1. DS3231: right-angle header, 32K removed, coin-cell charging resistor removed. BME280: 1 × 4 header, LED removed.
2. Thread the right tube. F2 on the right + plate, then solder the two right cell wires (step 2).
3. X9: each contact takes two wires (P2 + the continuation to X8), crimped together without folding back.
4. Right tube: twist SDA with GND and SCL with 3V3 (long I2C bus, next to SCK / BCLK).

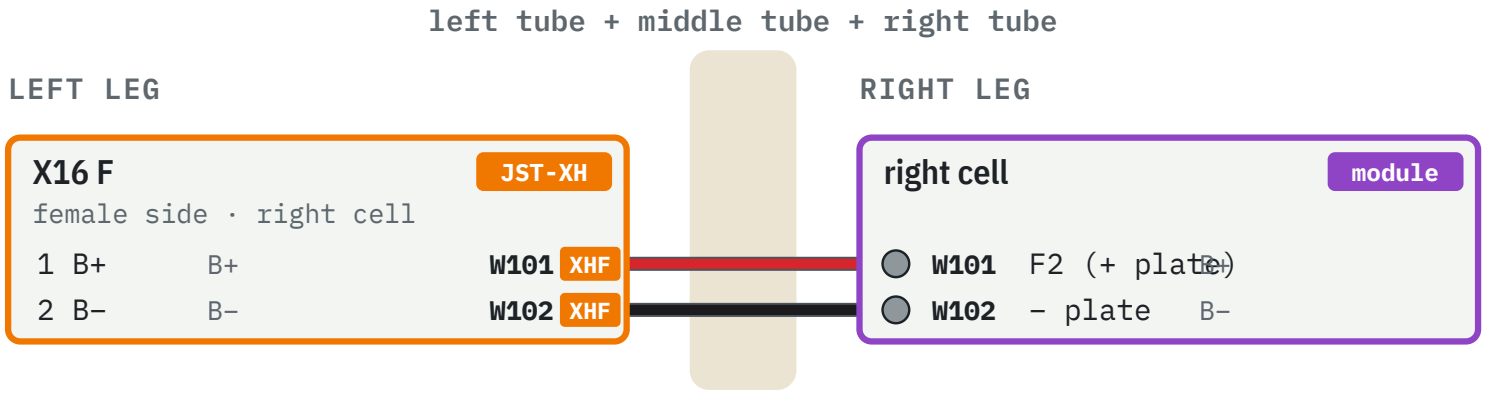
Bench test

- ☐ I2C scan: 0x53 (ADXL345), 0x57 (DS3231 EEPROM), 0x68 (DS3231), 0x76 or 0x77 (BME280).
- ☐ Before plugging in the right cell (X16): same voltage as the left one within 0.05 V.

Step 7 · Right tube and right leg: diagram 1/2: local wires, through right tube



Step 7 · Right tube and right leg: diagram 2/2: wires through left tube + middle tube + right tube



Wires finished at this step (10)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W101	red 26 AWG	B+	right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B−	left tube + middle tube + right tube	**52** → 55
W102	black 26 AWG	B−	right cell · − plate	XHF **X16** right cell · case 2 B− as B+; twisted with B+	left tube + middle tube + right tube	**57** → 60
W103	blue 30 AWG	GND	**P2** right middle distribution · hole C10 (wire 5 of the plan)	DF **X9** BME280 · case 2 GND	right tube	**27** → 30
W104	white 30 AWG	3V3	**P2** right middle distribution · hole C9 (wire 9 of the plan)	DF **X9** BME280 · case 1 VCC	right tube	**27** → 30
W105	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole C8 (wire 11 of the plan)	DF **X9** BME280 · case 4 SDA	right tube	**27** → 30
W106	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole C7 (wire 13 of the plan)	DF **X9** BME280 · case 3 SCL	right tube	**28** → 31
W107	blue 30 AWG	GND	DF **X9** BME280 · case 2 GND	DF **X8** DS3231 · case 6 GND double contact in X9 (with the wire from P2)	—	**14** → 17
W108	white 30 AWG	3V3	DF **X9** BME280 · case 1 VCC	DF **X8** DS3231 · case 5 VCC double contact in X9 (with the wire from P2)	—	**14** → 17
W109	green 30 AWG, 2 rings	SDA	DF **X9** BME280 · case 4 SDA	DF **X8** DS3231 · case 4 SDA double contact in X9 (with the wire from P2)	—	**14** → 17
W110	yellow 30 AWG, 2 rings	SCL	DF **X9** BME280 · case 3 SCL	DF **X8** DS3231 · case 3 SCL double contact in X9 (with the wire from P2)	—	**14** → 17

Wire lengths

Taken from the 3D model (tars4.scad): route from one end to the other along the column, through the pivot (z 238) when the wire changes column, +10 % for bends, + service loop at each end (7 cm at the P3 carrier, 4 cm at connectors, 2–3 cm at boards and modules), + 3 cm per tube (leg rotation). Length to cut = estimated + 3 cm; module pigtails are 6 cm. Cut at assembly time, after offering the wire up in TARS if possible.

colour	gauge	wires	to cut
yellow	30 AWG	11	3.61 m
green	30 AWG	12	3.49 m
brown	30 AWG	9	2.81 m
purple	30 AWG	7	2.49 m
blue	30 AWG	12	2.24 m
white	30 AWG	13	2.21 m
red	30 AWG	8	1.97 m
orange	30 AWG	5	1.46 m
red	26 AWG	15	3.32 m
black	26 AWG	10	2.49 m
blue	26 AWG	3	0.83 m
white	26 AWG	3	0.81 m
grey	26 AWG	2	0.28 m
total 30 AWG			20.28 m
total 26 AWG			7.73 m

Full wire list

step where it is started → finished

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W1	1	red 26 AWG, purple band	VUSB	**P4** power path · hole F8 (wire 1 of the plan)	TP4056 · IN+ (VUSB)	—	**10** → **13**
W2	1	red 26 AWG, white band	OUT+	**P4** power path · hole B4 (wire 2 of the plan)	TP4056 · OUT+	—	**11** → **14**
W3	1	black 26 AWG, white band	GND	**P4** power path · hole C1 (wire 7 of the plan)	TP4056 · OUT−	—	**11** → **14**
W4	1	red 30 AWG	B+	**P4** power path · hole J8 (wire 10 of the plan)	TP4056 · B+ (sense)	—	**11** → **14**
W5	1	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 5 of the plan)	TPS63020 · VIN	—	**12** → **15**
W6	1	black 26 AWG, orange band	GND	**P4** power path · hole D1 (wire 8 of the plan)	TPS63020 · GND	—	**12** → **15**
W7	1	white 30 AWG	3V3	**P4** power path · hole H8 (wire 13 of the plan)	TPS63020 · VOUT	—	**12** → **15**
W8	1	red 26 AWG, yellow band	VPATH	**P4** power path · hole A8 (wire 3 of the plan)	XHF **X10** J1 + SW1 · case 3 VPATH	—	**11** → **14**
W9	1	red 26 AWG, orange band	VSYS	**P4** power path · hole D3 (wire 4 of the plan)	XHF **X10** J1 + SW1 · case 4 VSYS	—	**10** → **13**
W10	1	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 6 of the plan)	XHF **X13** power block outputs · case 1 VSYS	—	**11** → **14**
W11	1	black 26 AWG, orange band	GND	**P4** power path · hole E1 (wire 9 of the plan)	XHF **X13** power block outputs · case 2 GND amp	—	**11** → **14**
W12	1	brown 30 AWG	VBAT/2	**P4** power path · hole J3 (wire 11 of the plan)	XHF **X14** measurements · case 1 VBAT/2	—	**10** → **13**
W13	1	orange 30 AWG	USB_DET	**P4** power path · hole E2 (wire 12 of the plan)	XHF **X14** measurements · case 2 USB_DET	—	**11** → **14**
W14	1	brown 30 AWG, 3 rings	SD_EN	**P4** power path · hole H5 (wire 15 of the plan)	XHF **X14** measurements · case 3 SD_EN	—	**9** → **12**
W15	1	red 26 AWG, purple band	VUSB	TP4056 · IN+	XHF **X10** J1 + SW1 · case 1 VUSB	—	**9** → **12**
W16	1	black 26 AWG, purple band	GND	TP4056 · IN−	XHF **X10** J1 + SW1 · case 2 GND	—	**9** → **12**
W17	1	red 26 AWG	B+	TP4056 · B+	XHM **X12** left cell · case 1 B+ cell junction: two wires soldered on the B+ pad	—	**10** → **13**
W18	1	black 26 AWG	B−	TP4056 · B−	XHM **X12** left cell · case 2 B− cell junction: two wires soldered on the B− pad	—	**10** → **13**

W19	**1**	red 26 AWG	B+	TP4056 · B+	XHM **X16** right cell · case 1 B+ to X16, parked above the power-path board	—	**13** → **16**
W20	**1**	black 26 AWG	B−	TP4056 · B−	XHM **X16** right cell · case 2 B−	—	**13** → **16**
W21	**1**	white 26 AWG	3V3	TPS63020 · VOUT	XHF **X13** power block outputs · case 3 3V3 along the logic GND wire	—	**11** → **14**
W22	**1**	blue 26 AWG	GND	TPS63020 · GND	XHF **X13** power block outputs · case 4 GND log logic GND, along the 3V3	—	**11** → **14**
W23	**2**	red 26 AWG, purple band	VUSB	J1 micro-USB · VBUS	XHM **X10** J1 + SW1 · case 1 VUSB twisted with the GND	—	**24** → **27**
W24	**2**	black 26 AWG, purple band	GND	J1 micro-USB · GND	XHM **X10** J1 + SW1 · case 2 GND	—	**24** → **27**
W25	**2**	red 26 AWG, yellow band	VPATH	SW1 switch · end lug	XHM **X10** J1 + SW1 · case 3 VPATH	—	**24** → **27**
W26	**2**	red 26 AWG, orange band	VSYS	SW1 switch · centre lug	XHM **X10** J1 + SW1 · case 4 VSYS	—	**24** → **27**
W27	**2**	red 26 AWG	B+	left cell · F1 (+ plate)	XHF **X12** left cell · case 1 B+ twisted with B−	—	**18** → **21**
W28	**2**	black 26 AWG	B−	left cell · − plate	XHF **X12** left cell · case 2 B−	—	**24** → **27**
W29	**3→4**	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → **31**
W30	**3→4**	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → **33**
W31	**3→4**	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → **29**
W32	**3→4**	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → **28**
W33	**3→4**	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → **30**
W34	**3→4**	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → **29**
W35	**3→4**	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → **39**
W36	**3→4**	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → **38**
W37	**3→4**	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → **25**
W38	**3→4**	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → **25**
W39	**3→4**	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → **25**
W40	**3→4**	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → **27**
W41	**3→4**	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	—	**22** → **25**
W42	**3→4**	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	—	**23** → **26**
W43	**4**	yellow 30 AWG, 1 ring	MIC_BCLK	INMP441 · SCK	DM **X3** INMP441 · case 1 SCK 5–6 cm pigtail	—	**6** → **7**

W44	**4**	purple 30 AWG, 1 ring	MIC_WS	● INMP441 · WS	DM **X3** INMP441 · case 2 WS	—	**6** → **7**
W45	**4**	green 30 AWG, 1 ring	MIC_SD	● INMP441 · SD	DM **X3** INMP441 · case 3 SD	—	**6** → **7**
W46	**4**	white 30 AWG	3V3	● INMP441 · VDD	DM **X3** INMP441 · case 4 VDD	—	**6** → **7**
W47	**4**	blue 30 AWG	GND	● INMP441 · GND	DM **X3** INMP441 · case 5 GND L/R bridged to GND on the board	—	**6** → **7**
W48	**4**	blue 30 AWG	GND	● **P1** left middle distribution · hole A10 (wire 1 of the plan)	DF **X1** screen 1 · case 1 GND	—	**15** → **18**
W49	**4**	white 30 AWG	3V3	● **P1** left middle distribution · hole A9 (wire 7 of the plan)	DF **X1** screen 1 · case 2 VCC	—	**15** → **18**
W50	**4**	brown 30 AWG	BL_EN	● **P1** left middle distribution · hole B8 (wire 13 of the plan)	DF **X1** screen 1 · case 8 BLK	—	**15** → **18**
W51	**4**	red 30 AWG	DC	● **P1** left middle distribution · hole B7 (wire 16 of the plan)	DF **X1** screen 1 · case 6 DC	—	**15** → **18**
W52	**4**	orange 30 AWG	RST	● **P1** left middle distribution · hole B6 (wire 19 of the plan)	DF **X1** screen 1 · case 5 RES	—	**15** → **18**
W53	**4**	green 30 AWG	MOSI	● **P1** left middle distribution · hole B5 (wire 22 of the plan)	DF **X1** screen 1 · case 4 SDA	—	**16** → **19**
W54	**4**	blue 30 AWG	GND	● **P1** left middle distribution · hole D11 (wire 6 of the plan)	DF **X3** INMP441 · case 5 GND	—	**16** → **19**
W55	**4**	white 30 AWG	3V3	● **P1** left middle distribution · hole D9 (wire 10 of the plan)	DF **X3** INMP441 · case 4 VDD	—	**15** → **18**
W56	**4**	white 30 AWG	3V3	DF **X3** INMP441 · case 4 VDD	DF **X2** ADXL345 · case 2 VCC double contact in X3 (with the wire from P1)	—	**12** → **15**
W57	**4**	blue 30 AWG	GND	DF **X3** INMP441 · case 5 GND	DF **X2** ADXL345 · case 1 GND double contact in X3 (with the wire from P1)	—	**12** → **15**
W58	**1→5**	white 30 AWG, 3 rings	SD_3V3	● **P4** power path · hole I6 (wire 14 of the plan)	DF **X15** microSD · case 4 3V3	—	**16** → **19**
W59	**3→5**	brown 30 AWG	VBAT/2	● **P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**49** → **52**
W60	**3→5**	orange 30 AWG	USB_DET	● **P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**52** → **55**
W61	**3→5**	brown 30 AWG, 3 rings	SD_EN	● **P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → **55**
W62	**3→5**	purple 30 AWG, 3 rings	SD_CMD	● **P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → **60**
W63	**3→5**	green 30 AWG, 3 rings	SD_D0	● **P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → **60**
W64	**3→5**	yellow 30 AWG, 3 rings	SD_CLK	● **P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → **58**
W65	**4→5**	blue 26 AWG	GND	● **P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → **36**

W66	**4→5**	white 26 AWG	3V3	**P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → **36**
W67	**4→5**	blue 30 AWG, 3 rings	GND	**P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → **38**
W68	**5**	yellow 30 AWG, 3 rings	SD_CLK	microSD reader · CLK	DM **X15** microSD · case 1 CLK 5–6 cm pigtail	—	**6** → **7**
W69	**5**	purple 30 AWG, 3 rings	SD_CMD	microSD reader · CMD	DM **X15** microSD · case 2 CMD	—	**6** → **7**
W70	**5**	green 30 AWG, 3 rings	SD_D0	microSD reader · D0	DM **X15** microSD · case 3 D0	—	**6** → **7**
W71	**5**	white 30 AWG, 3 rings	SD_3V3	microSD reader · VCC	DM **X15** microSD · case 4 3V3	—	**6** → **7**
W72	**5**	blue 30 AWG, 3 rings	GND	microSD reader · GND	DM **X15** microSD · case 5 GND	—	**6** → **7**
W73	**3→6**	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → **52**
W74	**3→6**	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → **54**
W75	**3→6**	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → **50**
W76	**3→6**	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → **49**
W77	**3→6**	green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	DF **X5** MAX98357A · case 3 DIN	middle tube	**57** → **60**
W78	**3→6**	purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	DF **X5** MAX98357A · case 1 LRC	middle tube	**58** → **61**
W79	**3→6**	yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	DF **X5** MAX98357A · case 2 BCLK	middle tube	**59** → **62**
W80	**4→6**	blue 30 AWG	GND	**P1** left middle distribution · hole C11 (wire 5 of the plan)	**P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → **32**
W81	**4→6**	white 30 AWG	3V3	**P1** left middle distribution · hole F9 (wire 11 of the plan)	**P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → **32**
W82	**4→6**	brown 30 AWG	BL_EN	**P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → **30**
W83	**4→6**	red 30 AWG	DC	**P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → **30**
W84	**4→6**	orange 30 AWG	RST	**P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → **30**
W85	**4→6**	green 30 AWG	MOSI	**P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → **31**
W86	**4→6**	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → **39**
W87	**4→6**	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → **40**

W88	**5→6**	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → **51**
W89	**5→6**	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B– pair in the tubes	left tube + middle tube	**48** → **51**
W90	**6**	white 30 AWG	PIR_VDD	● AM312 · VIN	DM **X7** AM312 · case 1 VIN 5–6 cm pigtail	–	**6** → **7**
W91	**6**	red 30 AWG, 1 ring	PIR_OUT	● AM312 · OUT	DM **X7** AM312 · case 2 OUT	–	**6** → **7**
W92	**6**	blue 30 AWG	GND	● AM312 · GND	DM **X7** AM312 · case 3 GND	–	**6** → **7**
W93	**6**	blue 30 AWG	GND	● **P2** right middle distribution · hole B10 (wire 4 of the plan)	DF **X4** screen 2 · case 1 GND	–	**14** → **17**
W94	**6**	white 30 AWG	3V3	● **P2** right middle distribution · hole B9 (wire 8 of the plan)	DF **X4** screen 2 · case 2 VCC	–	**15** → **18**
W95	**6**	brown 30 AWG, 1 ring	AMP_SD	● **P2** right middle distribution · hole F11 (wire 2 of the plan)	DF **X5** MAX98357A · case 5 SD	–	**16** → **19**
W96	**6**	blue 30 AWG	GND	● **P2** right middle distribution · hole D10 (wire 6 of the plan)	DF **X7** AM312 · case 3 GND	–	**14** → **17**
W97	**6**	white 30 AWG	PIR_VDD	● **P2** right middle distribution · hole D6 (wire 14 of the plan)	DF **X7** AM312 · case 1 VIN	–	**15** → **18**
W98	**6**	red 30 AWG, 1 ring	PIR_OUT	● **P2** right middle distribution · hole F4 (wire 16 of the plan)	DF **X7** AM312 · case 2 OUT	–	**17** → **20**
W99	**6**	grey 26 AWG	SPK+	● speaker · +	DF **X6** speaker · case 1 + + marked with a pen at both ends	–	**11** → **14**
W100	**6**	grey 26 AWG	SPK–	● speaker · –	DF **X6** speaker · case 2 –	–	**11** → **14**
W101	**2→7**	red 26 AWG	B+	● right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B–	left tube + middle tube + right tube	**52** → **55**
W102	**2→7**	black 26 AWG	B–	● right cell · – plate	XHF **X16** right cell · case 2 B– as B+; twisted with B+	left tube + middle tube + right tube	**57** → **60**
W103	**6→7**	blue 30 AWG	GND	● **P2** right middle distribution · hole C10 (wire 5 of the plan)	DF **X9** BME280 · case 2 GND	right tube	**27** → **30**
W104	**6→7**	white 30 AWG	3V3	● **P2** right middle distribution · hole C9 (wire 9 of the plan)	DF **X9** BME280 · case 1 VCC	right tube	**27** → **30**
W105	**6→7**	green 30 AWG, 2 rings	SDA	● **P2** right middle distribution · hole C8 (wire 11 of the plan)	DF **X9** BME280 · case 4 SDA	right tube	**27** → **30**
W106	**6→7**	yellow 30 AWG, 2 rings	SCL	● **P2** right middle distribution · hole C7 (wire 13 of the plan)	DF **X9** BME280 · case 3 SCL	right tube	**28** → **31**
W107	**7**	blue 30 AWG	GND	DF **X9** BME280 · case 2 GND	DF **X8** DS3231 · case 6 GND double contact in X9 (with the wire from P2)	–	**14** → **17**
W108	**7**	white 30 AWG	3V3	DF **X9** BME280 · case 1 VCC	DF **X8** DS3231 · case 5 VCC double contact in X9 (with the wire from P2)	–	**14** → **17**

W109 **7**	 green 30 AWG, 2 rings	SDA	DF **X9** BME280 · case 4 SDA	DF **X8** DS3231 · case 4 SDA double contact in X9 (with the wire from P2)	—	**14** → **17**
W110 **7**	 yellow 30 AWG, 2 rings	SCL	DF **X9** BME280 · case 3 SCL	DF **X8** DS3231 · case 3 SCL double contact in X9 (with the wire from P2)	—	**14** → **17**

CONNECTORS (HARNESS SIDE, CAVITY 1 ON THE LEFT)

X1 screen 1 · Dupont 1 × 8 on a right-angle header

GND	VCC	SCL	SDA	RES
GND	3V3	SCK1	MOSI	RST

DC	CS	BLK
DC	CS1	BL_EN

X2 ADXL345 · Dupont 1 × 8, 5 wires

GND	VCC	CS	INT1	INT2
GND	3V3	empty	INT1	plugged

SDO	SDA	SCL
empty	SDA	SCL

X3 INMP441 · pigtail + inline Dupont 1 × 6

SCK	WS	SD	VDD	GND
MIC_BCLK	MIC_WS	MIC_SD	3V3	GND

–
plugged

X4 screen 2 · Dupont 1 × 8 on a right-angle header

GND	VCC	SCL	SDA	RES
GND	3V3	SCK2	MOSI	RST

DC	CS	BLK
DC	CS2	BL_EN

X5 MAX98357A · Dupont 1 × 4 + 1 × 3

LRC	BCLK	DIN	GAIN	SD
LRC	BCLK	DIN	plugged	AMP_SD

GND	VIN
GND	VSYS

X6 speaker · Dupont 1 × 2 on a header

+	–
SPK+	SPK–

X7 AM312 · pigtail + inline Dupont 1 × 4

VIN	OUT	GND	–
PIR_VDD	PIR_OUT	GND	plugged

X8 DS3231 · Dupont 1 × 6 on a right-angle header, 4 wires

32K	SQW	SCL	SDA	VCC
plugged	empty	SCL	SDA	3V3

GND
GND

X9 BME280 · Dupont 1 × 4 (daisy-chained to X8)

VCC	GND	SCL	SDA
3V3	GND	SCL	SDA

X10 J1 + SW1 · inline JST-XH 4

VUSB	GND	VPATH	VSYS
VUSB	GND	VPATH	VSYS

X12 left cell · inline JST-XH 2

B+	B–
B+	B–

X13 power block outputs · inline JST-XH 4

VSYS	GND amp	3V3	GND log
VSYS	GND	3V3	GND

X14 measurements · inline JST-XH 3

VBAT/2	USB_DET	SD_EN
VBAT/2	USB_DET	SD_EN

X15 microSD · pigtail + inline Dupont 1 × 6

CLK	CMD	D0	3V3	GND
SD_CLK	SD_CMD	SD_D0	SD_3V3	GND

–
plugged

X16 right cell · inline JST-XH 2

B+	B–
B+	B–